UNIVERSITY OF MACAU
Faculty of Science and Technology
Department of Computer Science

From Prompting to Structured Alignment: Individual, Population, and Temporal Boundaries of LLM Survey Simulation

by

CHENYU YUAN

A thesis submitted in partial fulfillment of the requirements for the degree of

Master of Computer Science

Thesis Supervisor: PENGYANG WANG

April 2027

Abstract

University of Macau

From Prompting to Structured Alignment: Individual, Population, and Temporal Boundaries of LLM Survey Simulation

by

CHENYU YUAN

Thesis Supervisor: PENGYANG WANG

Large language models are increasingly used as virtual survey respondents, but most evaluations treat surveys as one-shot prompting tasks. This thesis presents a data-centric pipeline that reconstructs voter–candidate interactions from the American National Election Studies (ANES), aligns a compact 8B model with survey distributions through LoRA fine-tuning, and evaluates the resulting model at the individual, population, and temporal levels. On unseen respondents in the 2016–2019 waves, the fine-tuned model reaches a mean absolute error of 20.84, a correlation of 0.679, and a rating-induced choice accuracy of 0.518 against a chance level of 0.163. A gradient-boosted baseline on the same features reaches 20.58, 0.714, and 0.534; paired bootstrap shows no significant difference in MAE or choice, while the baseline has a higher correlation. On General Social Survey policy attitudes, fine-tuning reduces marginal and joint Jensen–Shannon divergence by 91% and 86% on the EDC configuration. The thesis then reports a sharp temporal boundary: individual-level correlation falls from 0.722 in 2019 to near zero in 2024, and both the language model and the tabular baseline collapse in 2024. Adding the 2022 wave to training does not help. The results suggest that structured alignment is necessary but not sufficient for social simulation, and that temporal distribution shift should be treated as a first-class evaluation axis.

Contents

Abstract ii

List of Abbreviations vii

1 Introduction 1
- 1.1 Related Work . . . 2

2 Data and Methodology 3
- 2.1 Data Construction . . . 3
- 2.2 Evaluation Protocols . . . 3
- 2.3 Fine-Tuning and Readouts . . . 4
- 2.4 Population-Level Pipeline . . . 4
- 2.5 Metrics . . . 4

3 Results 5
- 3.1 Individual-Level Results . . . 5
- 3.2 Population-Level Results . . . 5

4 Temporal Analysis and Ablations 7
- 4.1 Per-Year Decay . . . 7
- 4.2 The 2024 Collapse and a Failed Mitigation . . . 7
- 4.3 Ablations . . . 7

5 Limitations and Conclusion 9
- 5.1 Limitations . . . 9
- 5.2 Conclusion . . . 10

Bibliography 11

A Data Construction Details 13
- A.1 Data Audit . . . 13
- A.2 Splits . . . 13
- A.3 Sampling Strategies . . . 13
- A.4 Feature Definitions . . . 14

B Additional Results **15**
B.1 Per-Year PAS Results 15
B.2 Ablations 15
B.3 Cross-Pressure Subgroups 15

C FAS Details **16**
C.1 Full FAS Table 16
C.2 Prompt Templates 16
C.3 Hyperparameters 17

Vita **18**

List of Figures

4.1 Per-year correlation (a) and MAE (b) for the fine-tuned LLM and HistGB. Shaded bands are standard deviations over three seeds. 8

List of Tables

3.1 PAS in-distribution results on unseen respondents (2016–2019). Chance choice accuracy is 0.163. 6
3.2 FAS distributional alignment on GSS (JS divergence, lower is better). . 6

A.1 Raw and cleaned row counts. 13
A.2 Candidate splits. The two protocols are alternative designs and overlap across each other; within each protocol the splits are disjoint. 13

B.1 PAS by survey year on the held-out respondents. The 2022 and 2024 waves have two candidates, so chance choice accuracy is 0.5. 15
B.2 Ablations on the 2016–2019 in-distribution rows. 15
B.3 Cross-pressure subgroups on the 2016–2019 rows. 15

C.1 FAS distributional alignment on GSS (JS divergence, lower is better). Coverage is the fraction of the 400 requested rows returned. 16
C.2 Training and generation hyperparameters. 17

List of Abbreviations

ANES	American National Election Studies
ECE	Expected Calibration Error
FAS	Full Attribute Simulation
GSS	General Social Survey
IPF	Iterative Proportional Fitting
JS	Jensen–Shannon divergence
KL	Kullback–Leibler divergence
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MAE	Mean Absolute Error
PAS	Partial Attribute Simulation
RMSE	Root Mean Squared Error

CHAPTER 1:
Introduction

Questionnaire surveys are a primary source of evidence in the social sciences and in public policy, but they are expensive, slow, and difficult to field at scale. Large language models (LLMs) have been proposed as virtual survey respondents: given a demographic profile and a question, a model can produce an answer without recruiting a participant (Argyle et al., 2022; Dillion et al., 2023; Kim and Lee, 2023). Most existing work evaluates the model as a one-shot prompt. A profile is placed in the context window, a question is asked, and the answer is scored. This measures how well a model can role-play a respondent in a single turn. It does not test whether the model has internalized the structure of survey responses, whether its outputs are coherent at the population level, or whether the behaviour survives a change in the political environment.

This thesis takes a data-centric view. Survey responses are generated by a structured process: partisan identity, ideological position, and demographics interact with the attributes of a candidate, and the resulting evaluations follow regularities that are stable within an era but can shift across eras. If an LLM is to act as a credible respondent, it should be trained on that structure rather than prompted with it. The thesis reconstructs voter–candidate interactions from the American National Election Studies (ANES), aligns a compact 8B model with the resulting distribution through LoRA fine-tuning, and evaluates the model at the individual, population, and temporal levels.

The thesis is organized around three research questions. **RQ1**: can structured fine-tuning make a compact LLM a credible individual-level respondent, and how does it compare with a strong feature-engineered baseline? **RQ2**: can the same pipeline reproduce population-level survey distributions? **RQ3**: how does the behaviour transfer across survey waves? The thesis is therefore a boundary study: it asks when structured alignment helps, not whether a language model is universally better than existing methods.

The thesis makes four contributions. First, it provides a reproducible data construction and audit: trailing-period duplicates (8,165 rows) and group targets stored in the candidate column (23,060 rows) are removed, reducing 132,948 raw rows to 98,159 clean interaction-level rows, and two protocols (temporal and respondent-disjoint) respect the ANES panel. Second, on unseen respondents in 2016–2019, the fine-tuned model reaches an MAE of 20.84, a correlation of 0.679, and a choice accuracy of 0.518 against chance 0.163; a gradient-boosted baseline reaches 20.58, 0.714, and 0.534, and

paired bootstrap shows no significant MAE or choice difference. The value of the LLM is not superior point prediction; it is a unified interface that also produces a rating distribution and extends to population-level generation. Third, on GSS policy attitudes, fine-tuning reduces marginal and joint Jensen–Shannon divergence by 91% and 86% on EDC and by up to 76% and 64% on EHC, with supervised fine-tuning rather than retrieval or quota sampling as the driver. Fourth, across 2016–2024, individual-level correlation falls from 0.722 in 2019 to -0.056 in 2024, and the tabular baseline collapses in the same way; adding the 2022 wave does not help.

1.1 Related Work

Work on LLMs as virtual respondents conditions models on demographic backstories and compares the resulting "silicon samples" with survey margins (Argyle et al., 2022; Santurkar et al., 2023; Dillion et al., 2023; Kim and Lee, 2023); later work extends the paradigm to random silicon sampling, cross-survey probing, multi-domain benchmarks, and specialization to survey distributions (Sun et al., 2024; Zhou et al., 2024; Zhao et al., 2025; Cao et al., 2025). Most of this work evaluates a one-shot respondent and rarely separates individual accuracy from population-level alignment. A parallel literature documents the limits: models exhibit human-like response biases but also systematic deviations (Tjuatja et al., 2023), synthetic replacements can distort marginal and joint distributions (Bisbee et al., 2024), and the role of generative AI in social science remains contested (Grossmann et al., 2023; Bail, 2024; Ziems et al., 2024; Ma et al., 2024).

Persona fidelity and personalization have been studied extensively (Park et al., 2024; Wang et al., 2024; Xu et al., 2024; Salemi et al., 2024; Ng et al., 2024); that work concerns behavioural consistency within a conversation, whereas this thesis asks for statistical structure across a population. Fine-tuning has been used for preference alignment (Chu et al., 2023; Zhao et al., 2023); here the goal is competence alignment. For population-level generation, iterative proportional fitting (Deming and Stephan, 1940; Fienberg, 1970) and LOESS (Cleveland, 1979) provide the classical tools.

Prior work evaluates on a single wave or a pooled sample, which can hide temporal decay. The per-year analysis in Chapter 4 reports a monotonic decay for both a fine-tuned LLM and a tabular baseline on the same panel.

CHAPTER 2:
Data and Methodology

2.1 Data Construction

The individual-level corpus is built from the ANES pilot waves of 2016, 2018, 2019, 2020, 2022, and 2024 (132,948 rows, 3,165 respondents, 29 candidates). The population-level component uses the GSS waves of 2004–2022, with 2024 as the target year. The raw ANES table contains two defects. Trailing-period duplicates (`obama.` and `obama`, `hrc.` and `hrc`) produce 8,165 rows; social-group thermometers (`black`, `white`, `dem`, `rep`) stored in the candidate column produce 23,060 rows. Names are normalized, candidate and group targets are separated, missing or out-of-range scores are removed (3,588), and duplicates are dropped (8,141). The clean candidate table contains 98,159 rows.

Each observation is represented as a triple $(v_i, c_j, y_{ij} \mid t_i)$, where v_i is the voter profile, c_j the target, $y_{ij} \in [0, 100]$ the thermometer, and t_i the year. Candidate identity is decoupled from attributes: each candidate is described by a metadata vector $\mathscr{M}[c]$ containing party, ideology, age, position, and incumbent status. Features encode partisan, spatial, and conflict signals: party match ($+1$ co-partisan, -1 independent, 0 otherwise), seven-point ideological distance, a weighted similarity index (party 0.25, ideology 0.25, race 0.20, gender 0.15, age 0.15), and three cross-pressure indicators (same-party but distant; different-party but close; similar but different-party).

2.2 Evaluation Protocols

ANES is a panel: 3,080 of 3,165 respondents appear in more than one wave, and there are no test-only respondents in 2022/2024, so a temporal split measures transfer to a later wave, not generalization to unseen individuals. The temporal protocol (T) trains on 2016–2020 (6,516 rows), validates on 2022 (2,501), and tests on 2024 (3,421). The respondent protocol (R) splits respondents 80/20 by hash (5,094 train, 6,014 test rows); the strict T$\times$R cell contains 226 rows. R is the main in-distribution test; T measures the temporal boundary.

2.3 Fine-Tuning and Readouts

The model is Qwen3-8B. LoRA is applied to all attention and MLP projections with rank $r = 8$, scaling $\alpha = 32$, and dropout 0.1. Training uses three epochs, learning rate 10^{-4}, effective batch 16, and maximum length 512; the loss is on answer tokens only. This yields 21.8M trainable parameters (0.27%), with three seeds (42–44).

Generation decodes greedily and parses the first integer in $[0, 100]$; it preserves individual variation and is the primary choice readout. *Anchor likelihood* scores eleven anchors $a \in \{0, 10, \ldots, 100\}$ by sequence log-probability, normalizes with a softmax, and reports expectation and argmax. Choice selects the highest-scoring candidate within each respondent–year group and is compared with the rating-induced choice and chance.

2.4 Population-Level Pipeline

For full attribute simulation (FAS), every univariate and pairwise cell is extrapolated with a four-model ensemble: quadratic regression, LOESS with a tri-cube kernel, an exponential trend, and linear regression, weighted by 1.0, 1.0, 0.6, and 0.8 and normalized by 3.4. The predicted marginals and pairwise tables are combined with iterative proportional fitting: starting from the independence approximation, multiplicative updates match each pairwise table and marginal, run for 800 iterations, and are normalized with Laplace smoothing. The resulting joint distribution is used to generate five task types: batch generation (50%), prior-constrained sampling (25%), scenario variation (15%), subgroup analysis (7%), and iterative correction (3%).

Generation setup. All FAS samples were produced through a Dify application at a fixed endpoint, configured with an 8B Llama-family model (the project's GSS fine-tuned model for SFT, the base model otherwise); retrieval was toggled by switching application keys. Each run requested 20 rows per batch for 20 batches (400 rows) in one conversation. Outputs were parsed as JSON arrays of three values in $\{1, 2, 3\}$; failed batches were skipped, explaining the 351–400 row counts. Temperature is a fixed property of the Dify application. The quota condition used the same endpoint with 15 iterative correction rounds.

2.5 Metrics

PAS reports MAE, RMSE, Pearson correlation, relaxed accuracy (same 20-point bin or error ≤ 10), rating-induced choice accuracy, and ECE. FAS reports JS and symmetric KL on each marginal and pairwise joint, averaged over attributes or pairs.

CHAPTER 3:
Results

3.1 Individual-Level Results

The main in-distribution evaluation uses the respondent protocol restricted to the 2016, 2018, and 2019 waves: 2,726 rows from respondents never seen during training. Baselines are HistGB and Ridge on the same features, plus the base model with the same prompts but no fine-tuning. Confidence intervals are bootstrap 95% intervals over rows for MAE and correlation, and over respondent–year groups for choice.

Table 3.1 reports the result. The fine-tuned LLM reaches an MAE of 20.84 [20.38, 21.30], a correlation of 0.679 [0.666, 0.692], and a choice accuracy of 0.518 [0.481, 0.552] against chance 0.163. HistGB reaches 20.58 [20.01, 21.12], 0.714 [0.696, 0.733], and 0.534 [0.498, 0.569]; Ridge is weaker at 22.47, 0.655, and 0.484.

The honest reading is that the LLM is competitive with, but not superior to, a strong tabular baseline. A paired bootstrap over the same rows gives an LLM–GBM MAE difference of $+0.60$ $[-0.19, +1.42]$ for seed 42 (seeds 43 and 44: $+0.02$ and $+0.19$), and a choice difference of $+0.026$ $[-0.007, +0.062]$; both intervals include zero. The correlation difference is -0.039 $[-0.056, -0.021]$, consistently negative across seeds. Fine-tuning, by contrast, improves MAE by 2.98 $[2.10, 3.82]$ relative to the base model, with a correlation gain of $+0.082$ and a choice gain of $+0.081$. The language model therefore adds a unified interface and a rating distribution, not superior point prediction.

Predictions are correlated with the truth but compressed toward the middle of the scale, which motivates reporting prediction spread alongside calibration. The per-year breakdown is reported in Chapter 4.

3.2 Population-Level Results

For each GSS configuration (EDC and EHC), the 2024 joint distribution is reconstructed from eight historical waves, 400 synthetic respondents are sampled, and the sample is compared with the empirical 2024 distribution. JS divergence is reported on the three marginals and the three pairwise joints; lower is better.

On EDC, the base model produces a marginal JS of 0.0552 and a joint JS of 0.1181. Direct SFT reduces these to 0.0048 and 0.0160 (91% and 86%), and the best run (v3) reaches 0.0040 and 0.0215 (93% and 82%). On EHC, the base model starts at 0.0319 and 0.1001; direct SFT reaches 0.0167 and 0.0470 (48% and 53%), and v3 reaches

Model	MAE	Correlation	Choice
Fine-tuned LLM (3 seeds)	20.84	0.679	0.518
HistGB	20.58	**0.714**	0.534
Ridge	22.47	0.655	0.484
Base LLM	24.16	0.593	0.479

Table 3.1: PAS in-distribution results on unseen respondents (2016–2019). Chance choice accuracy is 0.163.

Config	Run	n	Marginal JS	Joint JS
EDC	Base	400	0.0552	0.1181
	SFT	381	0.0048	0.0160
	SFT (v3)	400	**0.0040**	0.0215
EHC	Base	351	0.0319	0.1001
	SFT	400	0.0167	0.0470
	SFT (v3)	400	**0.0077**	**0.0363**

Table 3.2: FAS distributional alignment on GSS (JS divergence, lower is better).

0.0077 and 0.0363 (76% and 64%). EDC therefore supports the claim that fine-tuning substantially improves population-level alignment, including pairwise structure. EHC is weaker, and the "80% reduction" claim is scoped to EDC.

Component analysis shows that RAG alone is close to the base model (EDC 0.0473, EHC 0.0404) and that RAG plus SFT performs like direct SFT. The gain comes from supervised exposure to the reconstructed distribution and the five task types, not from retrieval. Quota annealing (EDC 0.0091, EHC 0.0175) is worse than direct SFT and is not used.

The base model returns 351 of 400 rows on EHC (coverage 0.877), while the SFT runs return 399–400. Charging every missing row as maximally wrong raises the base model's EHC marginal/joint JS from 0.0319/0.1001 to 0.1168/0.1850, while the fully covered v3 run stays at 0.0077/0.0363; on EDC, direct SFT's 0.0048/0.0160 becomes 0.0377/0.0489, whereas v3 is unaffected at 0.0040/0.0215. The v3 run is therefore the cleanest headline result. On EHC the joint symmetric KL increases from 0.4289 to 0.5487 for direct SFT even though JS falls, so JS is the primary metric.

CHAPTER 4:
Temporal Analysis and Ablations

4.1 Per-Year Decay

The temporal protocol trains on 2016–2020 and tests on 2024. Figure 4.1 and the full table in Appendix B report a per-year view of the respondent protocol: for each wave from 2016 to 2024, the model is evaluated on the rows contributed by the 20% held-out respondents. This holds respondent identity fixed while letting the year vary.

For the fine-tuned LLM, the correlation is 0.722 in 2019, 0.478 in 2020, 0.408 in 2022, and -0.056 in 2024; MAE rises from 19.20 to 25.39, 27.76, and 37.30. HistGB decays later but collapses in the same way: 0.746, 0.560, 0.657, and -0.008; MAE 18.85, 23.41, 27.98, and 35.52. The decay begins between 2019 and 2020, not at 2024. The LLM decays earlier than the tabular baseline: by 2022 its correlation is 0.408 while HistGB is still at 0.657.

4.2 The 2024 Collapse and a Failed Mitigation

On the full 2024 test set the fine-tuned LLM reaches an MAE of 36.47 ± 0.75 but a correlation of -0.029 ± 0.021; HistGB reaches 35.06 and -0.030. The predictions concentrate on the middle of the scale: 3,173 of 3,421 LLM predictions are exactly 50, and every Biden row is predicted as 50 (standard deviation 0.0), while Trump rows have a standard deviation of 17.4 and a correlation of -0.014. The anchor readout does not rescue the signal: its correlation is 0.003 with a prediction standard deviation of 2.1, so its low ECE is an artefact of near-constant predictions. This is not LLM-specific: HistGB, which has no language-model component, also loses the individual signal.

A natural mitigation is to add the most recent wave to training. The model is fine-tuned on the union of the 2016–2020 pool and the 2022 validation rows (9,017 rows) and evaluated on 2024. The result is unchanged: MAE 36.62, correlation -0.030, choice 0.525. Adding 2022 does not restore the signal, which points to a genuine distribution shift rather than a lack of recent data.

4.3 Ablations

Three design choices are isolated with matched runs. A1 replaces the nine-strategy pool with a uniform random sample of the same size. A2 removes the interaction block from

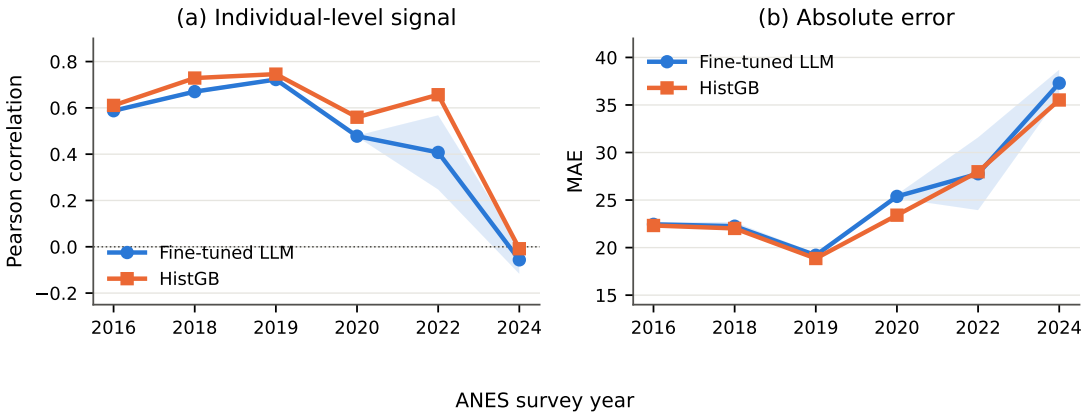


Figure 4.1: Per-year correlation (a) and MAE (b) for the fine-tuned LLM and HistGB. Shaded bands are standard deviations over three seeds.

the prompt. A3 additionally removes the candidate's party, position, and age. All use seed 42 and the same in-distribution rows.

The full model reaches an MAE of 20.84 $\pm$ 0.24, a correlation of 0.679 $\pm$ 0.003, and a choice accuracy of 0.553 $\pm$ 0.011. A1 reaches 21.27, 0.658, and 0.543; A2 reaches 21.13, 0.670, and 0.567; A3 reaches 21.45, 0.657, and 0.565. The direction is consistent but the effects are small: removing the distribution-aware sampling or the candidate metadata worsens MAE by roughly half a point and lowers correlation by about 0.02. The thesis does not claim that each component is individually necessary; the full table is in Appendix B.

The cross-pressure analysis gives a finer-grained view. The fine-tuned LLM is substantially better than HistGB on CP3, the identity-versus-party conflict (MAE 15.51–17.80 versus 20.31), and comparable on CP1 (18.77–21.55 versus 21.55). It is slightly worse on CP2 (23.21–24.13 versus 23.18) and on non-conflict rows (20.45–21.09 versus 20.14). The cells are small, so the result is descriptive.

CHAPTER 5:
Limitations and Conclusion

5.1 Limitations

Model and survey coverage. All fine-tuning uses Qwen3-8B; a second family was not evaluated because Llama-3.1-8B is gated. The pipeline is model-agnostic, but the claims are conditional on Qwen3-8B. Both ANES and GSS are U.S. surveys; cross-national, non-English, and non-political domains are not tested.

Group attitudes. A nine-group corpus was constructed but not evaluated; coverage is uneven across waves (three groups in 2016, nine in 2020, two in 2022), so the claims are restricted to candidate PAS and GSS FAS.

Panel and choice. ANES is a panel with no test-only respondents in 2022/2024, and the T$\times$R cell is 226 rows. Choice is rating-induced, not observed vote.

Ablations and FAS. Ablations use one seed; effects are small. FAS covers two configurations; EHC is weaker and its joint symmetric KL moves opposite for direct SFT. Temporal extrapolation is validated only for 2024, and only one mitigation is tested.

Pretraining contamination. Qwen3-8B and the Llama-family models used for FAS may have seen ANES/GSS data, election coverage, or published poll results during pretraining. The in-distribution signal may therefore reflect memorization as well as learned structure, and the 2024 collapse may be shaped by knowledge cutoff or contamination. The experiments cannot rule this out; the claims should be read as the behaviour of these models, not as learning from scratch.

FAS uncertainty. FAS metrics are computed on a single 400-row generation per condition; no confidence intervals or repeated generations are reported. Differences between adjacent conditions (e.g., 91% vs 93%) should therefore be read as descriptive.

Ethics. Synthetic respondents are not a substitute for real participants. A model trained on historical data can reproduce and amplify its biases, and the temporal results warn that an in-distribution simulator can lose all individual signal four years later. Deployments should report the training era, validation protocol, model family, and generated-sample coverage, and synthetic results should be clearly labelled.

5.2 Conclusion

This thesis presented a data-centric pipeline for LLM survey simulation and evaluated it at the individual, population, and temporal levels. The pipeline reconstructs interaction-level ANES data, decouples candidate identity from attributes, encodes partisan and ideological structure as explicit features, samples the training pool to cover tails and conflict cases, and fine-tunes a compact 8B model with LoRA; the same pipeline reconstructs a GSS joint distribution and fine-tunes a model to generate a synthetic population.

The individual-level result is deliberately modest. On unseen respondents in the same era, the fine-tuned model reaches an MAE of 20.84, a correlation of 0.679, and a choice accuracy of 0.518 against chance 0.163. A gradient-boosted baseline reaches 20.58, 0.714, and 0.534, and paired bootstrap shows no significant MAE or choice difference. The value of the language model is not superior point prediction; it is a unified interface that produces a rating distribution and extends to population-level generation. The population-level result is stronger for EDC: fine-tuning reduces marginal and joint JS divergence by 91% and 86% (93% and 82% for the fully covered run).

The temporal result is the most important negative finding. Individual-level correlation falls from 0.722 in 2019 to -0.056 in 2024, and the tabular baseline collapses in the same way. Adding the most recent wave does not help. Structured alignment is necessary but not sufficient for temporal generalization, and temporal shift should be treated as a first-class evaluation axis. Future work should replicate the pipeline on a second model family, extend it to cross-national and group-attitude data, develop temporal adaptation methods, and validate rating-induced choice against observed vote choice.

Bibliography

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J., Rytting, C., and Wingate, D. (2022). Out of one, many: Using language models to simulate human samples.

Bail, C. A. (2024). Can generative AI improve social science? *Proceedings of the National Academy of Sciences*, 121(21):e2314021121.

Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B., and Larson, J. M. (2024). Synthetic replacements for human survey data? the perils of large language models.

Cao, Y., Liu, H., Arora, A., Augenstein, I., Röttger, P., and Hershcovich, D. (2025). Specializing large language models to simulate survey response distributions for global populations. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics*.

Chu, E., Andreas, J., Ansolabehere, S., and Roy, D. (2023). Language models trained on media diets can predict public opinion.

Cleveland, W. S. (1979). Robust locally weighted regression and smoothing scatterplots. *Journal of the American Statistical Association*, 74(368):829–836.

Deming, W. E. and Stephan, F. F. (1940). On a least squares adjustment of a sampled frequency table when the expected marginal totals are known. *Annals of Mathematical Statistics*, 11(4):427–444.

Dillion, D., Tandon, N., Gu, Y., and Gray, K. (2023). Can AI language models replace human participants? *Trends in Cognitive Sciences*, 27(7):597–600.

Fienberg, S. E. (1970). An iterative procedure for estimation in contingency tables. *Annals of Mathematical Statistics*, 41(3):907–917.

Grossmann, I., Feinberg, M., Parker, D. C., Christakis, N. A., Tetlock, P. E., and Cunningham, W. A. (2023). AI and the transformation of social science research. *Science*, 380(6650):1108–1109.

Kim, J. and Lee, B. (2023). AI-augmented surveys: Leveraging large language models and surveys for opinion prediction.

Ma, B., Wang, X., Hu, T., Haensch, A.-C., Hedderich, M. A., Plank, B., and Kreuter, F. (2024). The potential and challenges of evaluating attitudes, opinions, and values in large language models.

Ng, M. T., Tse, H. T., Huang, J.-t., Li, J., Wang, W., and Lyu, M. R. (2024). How well can LLMs echo us? evaluating AI chatbots' role-play ability with ECHO.

Park, J. S., Zou, C. Q., Shaw, A., Hill, B. M., Cai, C. J., Morris, M. R., Willer, R., Liang, P., and Bernstein, M. S. (2024). Generative agent simulations of 1,000 people.

Salemi, A., Mysore, S., Bendersky, M., and Zamani, H. (2024). LaMP: When large language models meet personalization. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., and Hashimoto, T. (2023). Whose opinions do language models reflect? In *Proceedings of the 40th International Conference on Machine Learning*.

Sun, S., Lee, E., Nan, D., Zhao, X., Lee, W., Jansen, B. J., and Kim, J.-H. (2024). Random silicon sampling: Simulating human sub-population opinion using a large language model based on group-level demographic information.

Tjuatja, L., Chen, V., Wu, T., Talwalkar, A., and Durmus, E. (2023). Do LLMs exhibit human-like response biases? a case study in survey design.

Wang, X., Xiao, Y., Huang, J.-t., Yuan, S., Xu, R., Guo, H., Tu, Q., Fei, Y., Leng, Z., Wang, W., et al. (2024). InCharacter: Evaluating personality fidelity in role-playing agents through psychological interviews. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Xu, R., Wang, X., Chen, J., Yuan, S., Yuan, X., Liang, J., Chen, Z., Dong, X., and Xiao, Y. (2024). Character is destiny: Can large language models simulate persona-driven decisions in role-playing?

Zhao, J., Yuan, C., Luo, W., Xie, H., Zhang, G., Quan, S. J., Yuan, Z., Wang, P., and Zhang, D. (2025). Large language models as virtual survey respondents: Evaluating sociodemographic response generation.

Zhao, S., Dang, J., and Grover, A. (2023). Group preference optimization: Few-shot alignment of large language models.

Zhou, M., Yu, L., Geng, X., and Luo, L. (2024). ChatGPT vs social surveys: Probing the objective and subjective human society.

Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., and Yang, D. (2024). Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291.

APPENDIX A:
Data Construction Details

A.1 Data Audit

Table A.1 reports the raw and cleaned row counts and the main data defects. The trailing-period duplicates are normalized before deduplication; the group-like targets are removed from the candidate table and kept in the group corpus. The cleaned table retains the original row index so that every row can be traced back to the raw ANES table.

Item	Count
Raw rows	132,948
Trailing-period duplicate rows	8,165
Group targets mixed into candidates	23,060
Missing or out-of-range scores	3,588
Duplicate rows removed after normalization	8,141
Clean rows	98,159

Table A.1: Raw and cleaned row counts.

A.2 Splits

Split	Years	Rows	Respondents
`train_t`	2016–2020	6,516	all
`val_t`	2022	2,501	sampled
`test_t`	2024	3,421	1,712
`train_r`	2016–2020	5,094	80%
`test_r`	2016–2022	6,014	192
`test_tr`	2024	226	113

Table A.2: Candidate splits. The two protocols are alternative designs and overlap across each other; within each protocol the splits are disjoint.

A.3 Sampling Strategies

The training pool is drawn with nine complementary strategies: balanced coverage of the score range, hard cases near the middle of the scale, party-by-ideology cells,

similarity-stratified bins, cross-pressure cases, extreme scores, incumbent effects, temporal coverage, and candidate-specific balance. The pool contains 6,516 rows and is intentionally not a uniform sample; the test sets are built from all cleaned rows in the target years so that evaluation is not shaped by the same sampling design.

A.4 Feature Definitions

The core features are party match (+1 same party, -1 independent, 0 otherwise), seven-point ideological distance, a composite demographic similarity index, and three cross-pressure indicators: CP1 (same party, ideological distance ≥ 3), CP2 (different party, distance ≤ 1), and CP3 (similarity ≥ 0.7, different party).

APPENDIX B:

Additional Results

B.1 Per-Year PAS Results

		Fine-tuned LLM			HistGB		
Year	n	MAE	corr	choice	MAE	corr	choice
2016	697	22.45	0.588	0.329	22.32	0.611	0.256
2018	727	22.25	0.670	0.530	22.02	0.729	0.544
2019	1,302	19.20	0.722	0.663	18.85	0.746	0.639
2020	2,868	25.39	0.478	0.326	23.41	0.560	0.278
2022	194	27.76	0.408	0.674	27.98	0.657	0.804
2024	226	37.30	−0.056	0.448	35.52	−0.008	0.460

Table B.1: PAS by survey year on the held-out respondents. The 2022 and 2024 waves have two candidates, so chance choice accuracy is 0.5.

B.2 Ablations

Configuration	MAE	Correlation	Choice
Full (9-strategy)	20.84 ± 0.24	0.679 ± 0.003	0.553 ± 0.011
A1 random sampling	21.27	0.658	0.543
A2 minimal prompt	21.13	0.670	0.567
A3 name-only prompt	21.45	0.657	0.565

Table B.2: Ablations on the 2016–2019 in-distribution rows.

B.3 Cross-Pressure Subgroups

Subgroup	LLM MAE (3 seeds)	HistGB MAE	n
CP1 party vs. ideology	18.77–21.55	21.55	154
CP2 ideology vs. party	23.21–24.13	23.18	318
CP3 identity vs. party	15.51–17.80	20.31	164
No conflict	20.45–21.09	20.14	2,090

Table B.3: Cross-pressure subgroups on the 2016–2019 rows.

APPENDIX C:
FAS Details

C.1 Full FAS Table

Config	Run	n	Coverage	Marginal JS	Joint JS
EDC	Base	400	1.000	0.0552	0.1181
	SFT	381	0.953	0.0048	0.0160
	SFT (v3)	400	1.000	**0.0040**	0.0215
	RAG	376	0.940	0.0473	0.1096
	RAG + SFT	381	0.953	0.0054	0.0176
	Quota	395	0.988	0.0091	0.0216
EHC	Base	351	0.877	0.0319	0.1001
	SFT	400	1.000	0.0167	0.0470
	SFT (v3)	400	1.000	**0.0077**	**0.0363**
	RAG	379	0.948	0.0404	0.1095
	RAG + SFT	380	0.950	0.0152	0.0427
	Quota	400	1.000	0.0175	0.0522

Table C.1: FAS distributional alignment on GSS (JS divergence, lower is better). Coverage is the fraction of the 400 requested rows returned.

C.2 Prompt Templates

For PAS, the system message defines the task and fixes the output format; the user message contains the voter profile, the candidate block, and the interaction block. A representative user message is:

> Predict the support score (0–100) for this voter–candidate pair.
>
> Voter Profile: 45-year-old Female White; Bachelor's degree; Strong Democrat, Liberal.
>
> Candidate: Biden; Democrat; President; Age 78.
>
> Key Interactions: Same party; Ideology distance 0.5; Age difference 33 years; Year 2020.
>
> Support score (0–100):

For FAS, the model is given the target year, the three attributes and their levels, and the task type (batch generation, prior sampling, scenario variation, subgroup analysis, or iterative correction). The expected output is a list of rows, one per synthetic respondent.

C.3 Hyperparameters

Parameter	Value
Base model	Qwen3-8B
LoRA rank / alpha / dropout	8 / 32 / 0.1
Epochs	3
Learning rate	10^{-4}
Effective batch size	16
Maximum sequence length	512
Seeds	42, 43, 44
IPF iterations	800
FAS target samples	400

Table C.2: Training and generation hyperparameters.

Vita

CHENYU YUAN received the degree of Master of Computer Science from the University of Macau in April 2027. Her research interests include large language models, computational social science, and survey simulation.